An Intelligent Machine Learning–Based Intrusion Detection System for Ransomware Detection

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PROGRESS THROUGH KNOWLEDGE

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ABSTRACT

- Ransomware attacks pose a serious cyber security threat, causing data breaches and financial losses. Traditional signature-based intrusion detection systems are ineffective against evolving ransomware attacks. This paper proposes an intelligent machine learning–based intrusion detection stem that uses behavioural analysis of network and system activities to detect ransomware.

- Features such as abnormal file access and unusual network connections are used to train SVM, XGBoost and Naïve Bayes. The system enables early detection with improved accuracy and reduced false positives, providing an adaptive and reliable solution for ransomware protection.

INTRODUCTION

- **Ransomware Threat Overview:** Ransomware is a type of malware that encrypts files or locks systems and demands a ransom payment (usually in cryptocurrency) to restore access.
- **Behavioural Analysis for Ransomware Detection:** The system detect ransomware using behavioural patterns instead of signatures. It monitors abnormal file unauthorised operations, and unusual network connections**.**
- **Machine Learning Classification: XGBoost**, SVM, and Decision Tree models classify activities as normal or ransomware-related. Machine learning enables adaptive detection of evolving ransomware attacks**.**
- **Early Detection and System Reliability:** The system detects ransomware at an early stage, minimizing data and financial loss. It achieves high accuracy with reduced false positive providing a reliable security solution..

OBJECTIVE

- To detect ransomware using XGBoost Machine Learning model
- To monitor file system activities in real time
- To automatically quarantine malicious files
- To generate alerts and maintain security logs

LITERATURE SURVEY

S.No	Paper Title	Author name	Remarks
1	Exploring Ransomware Detection MayurRele Based on Artificial Intelligence and JohnSamuel Machine LearningDi pti Patil	Udaya Krishnan	Proposes an AI and machine learning– based Ransomware detection approach that combines anomaly detection and classification with feature extraction from system logs, network traffic, and file metadata.
2	Ransomware Detection Using Machine Learning	K.Razak,S. Abdullah, and M. Ibrah im Alraizza, Algarni	An intelligent machine learning–based intrusion detection approach for ransomware attacks, highlighting how ML models can effectively analyze system .
3.	Behavior-Based Ransomware Detection Using Supervised MachineLearning Algorithms.	R. Kumarand A. Sharma	This research emphasizes behavior-based ransomware detection using supervised ML algorithms. It demonstrates how monitoring system behavior improves early detection and resilience against zero- day ransomware attacks.

EXISTING SYSTEM

Existing System:

- Signature-based antivirus detection
- Manual scanning and updates
- Limited real-time monitoring

Disadvantages:

- Cannot detect zero-day attacks
- No machine learning intelligence
- Delayed detection
- No automatic quarantine

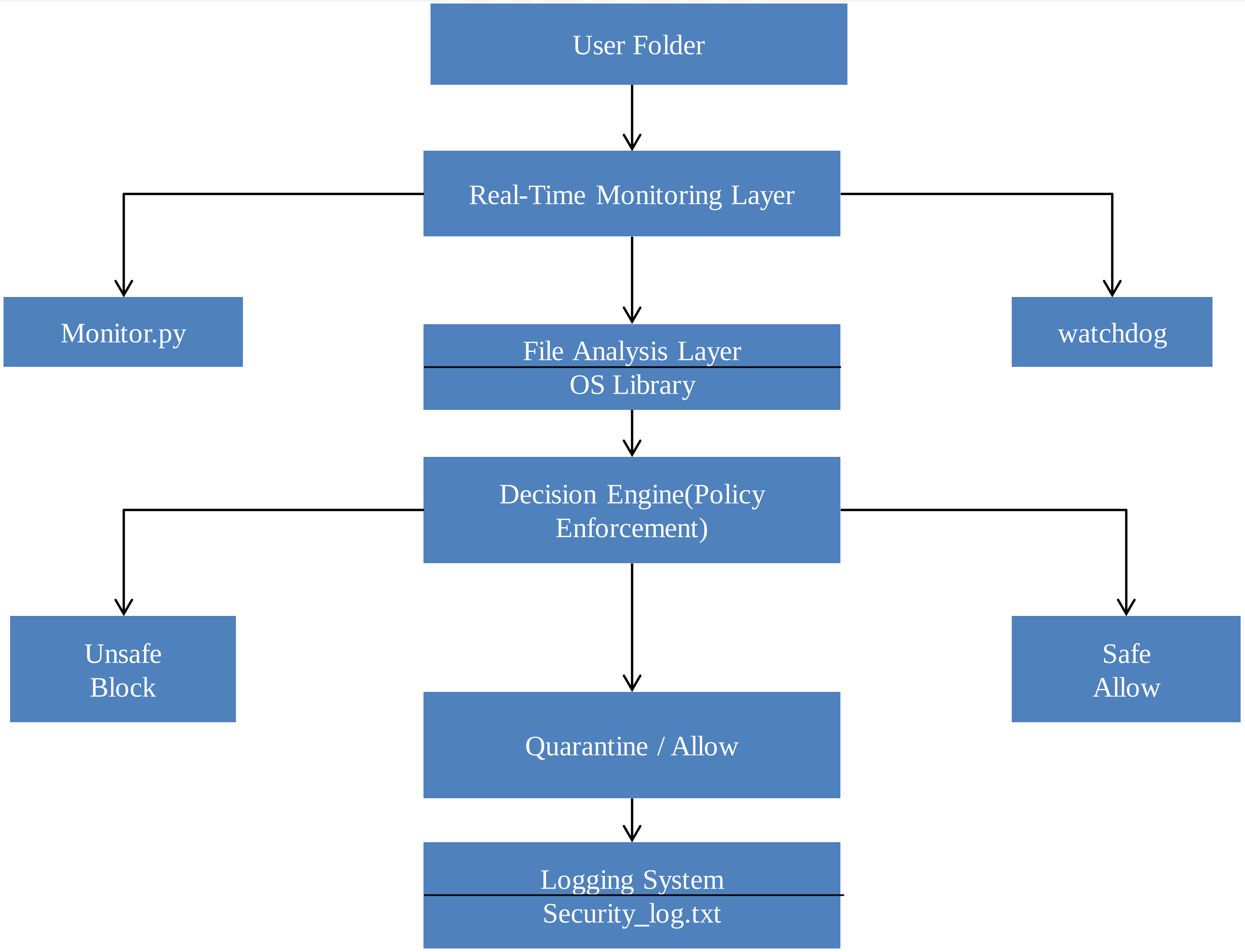
PROPOSED SYSTEM

- Real-time file monitoring
- XGBoost-based ransomware detection
- Probability-based classification
- Automatic quarantine
- Security logging

Advantages :

- Detects unknown ransomware
- Immediate blocking
- Reduced manual effort
- Intelligent ML-based approach

ARCHITECTURE



SOFTWARE REQUIREMENTS

- Programming Language : Python 3.10+
- Frontend : HTML, CSS, JavaScript
- Backend : Python
- ML Libraries : Scikit-learn, XGBoost
- Monitoring Library : Watchdog
- Dataset Format : CSV
- IDE : VS Code / PyCharm
- OS : Windows 10/11
- Browser : Chrome / Edge

MODULES

- Module 1: Data Collection & Preprocessing
- Module 2: Machine Learning Model Training
- Module 3: Real-Time File Monitoring
- Module 4: Ransomware Detection & Response
- Module 5: Reporting & Alert System

Module 1: Data Collection & Preprocessing

Module Description :

- This module focuses on preparing the ransomware dataset and training the Machine Learning model for classification.

Process:

- Collected ransomware dataset in CSV format
- Performed data preprocessing (cleaning, handling missing values)
- Separated features (X) and target label (y)
- Applied feature scaling (if required)
- Trained the XGBoost classification model
- Evaluated model accuracy
- Saved the trained model as xgb_model.pkl

Algorithm Used: XGBoost Classifier

Module 2: Machine Learning Model

Module Description :

This module continuously monitors a specified folder for newly created files.

Process:

- Configured monitoring folder (Secure Watch)
- Used Watchdog library to observe file system events
- Detected newly created files instantly
- Identified file extension

Technology Used:

- Python
- Watchdog Library

Module 3:Real-Time File Monitoring

Description:

This module performs intelligent ransomware detection using the trained XGBoost model.

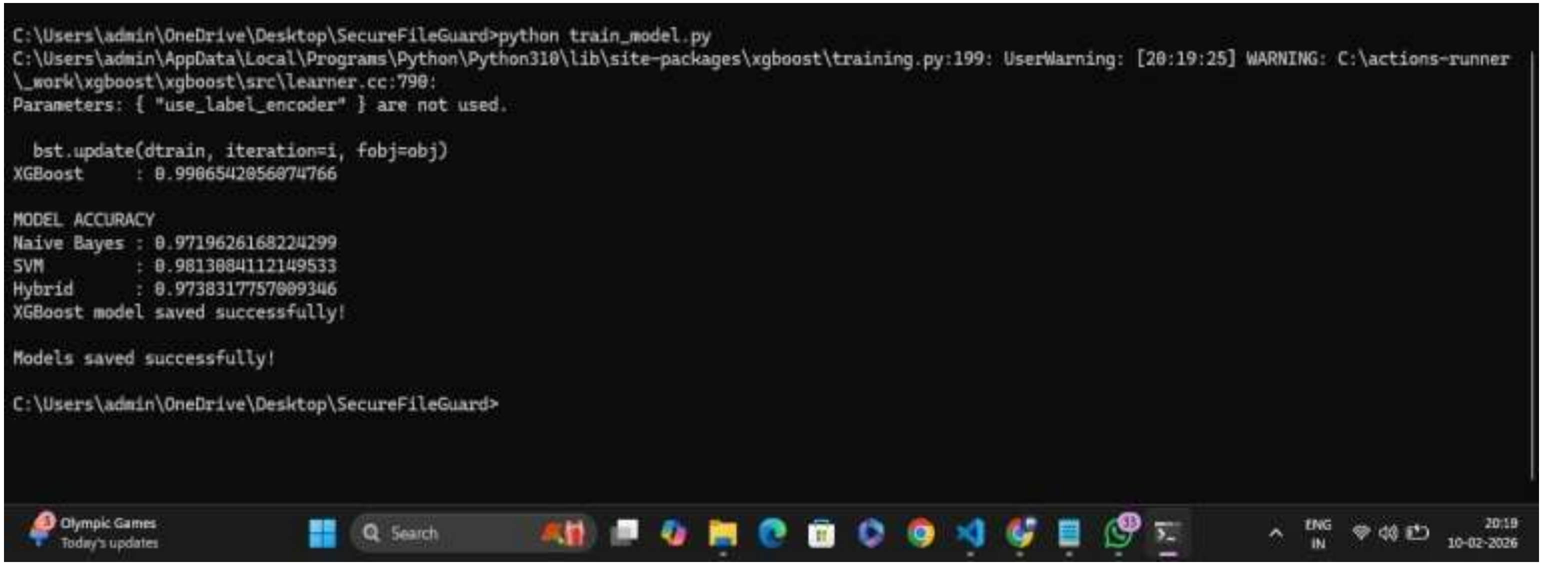
Process:

- Extracted file features
- Passed features to trained ML model
- Calculated malicious probability
- Compared probability with threshold
- Classified file as Safe or Malicious
- Automatically quarantined malicious files

Algorithm Used:

XGBoost Probability Prediction

Module 1 : Output Screenshot



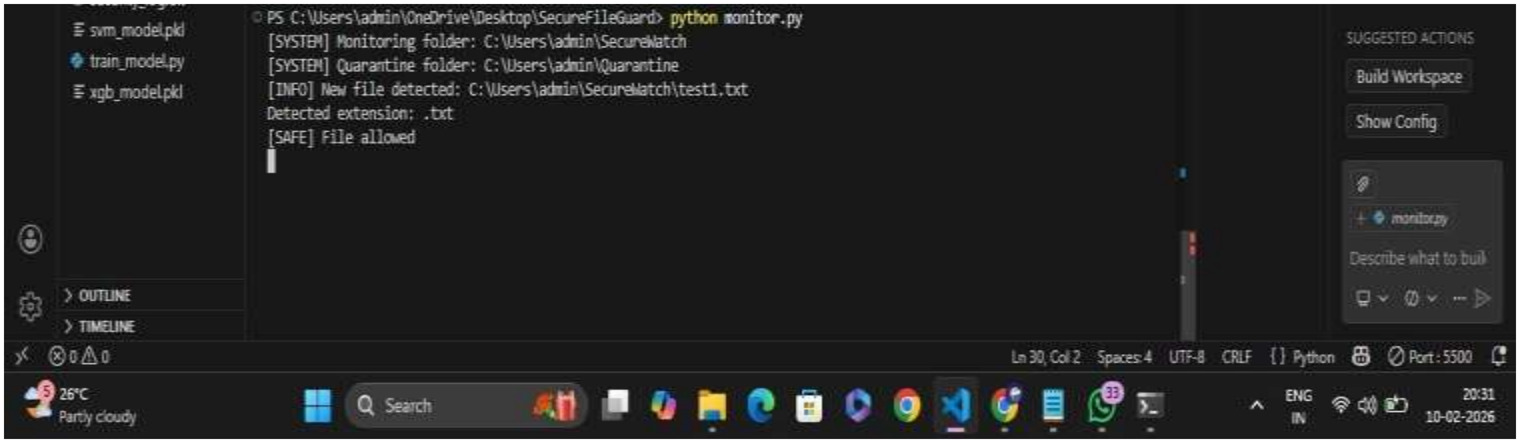
MODULE OUTPUT

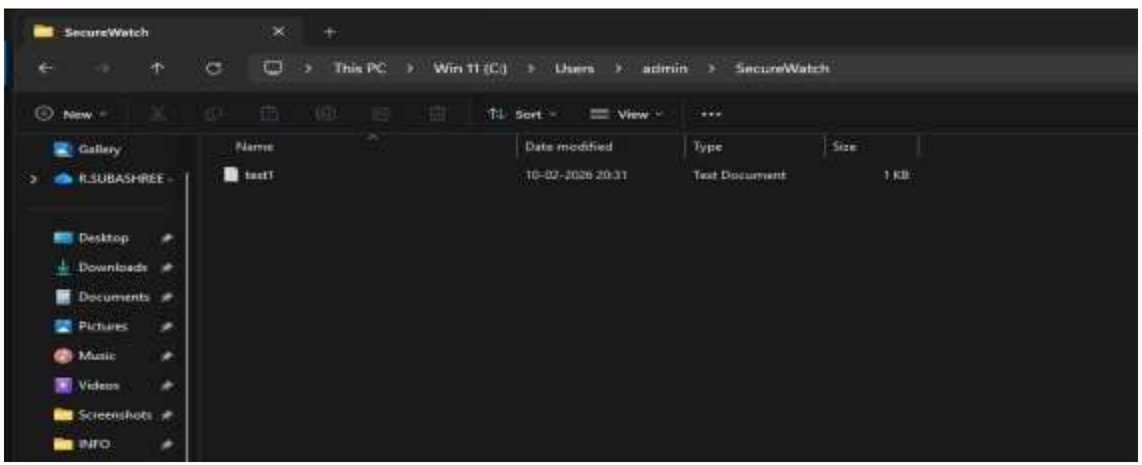
MODULE OUTPUT :

```
PS C:\Users\admin\OneDrive\Desktop\SecureFileGuard> python monitor.py
[SYSTEM] Monitoring folder: C:\Users\admin\SecureWatch
[SYSTEM] Quarantine folder: C:\Users\admin\Quarantine
```

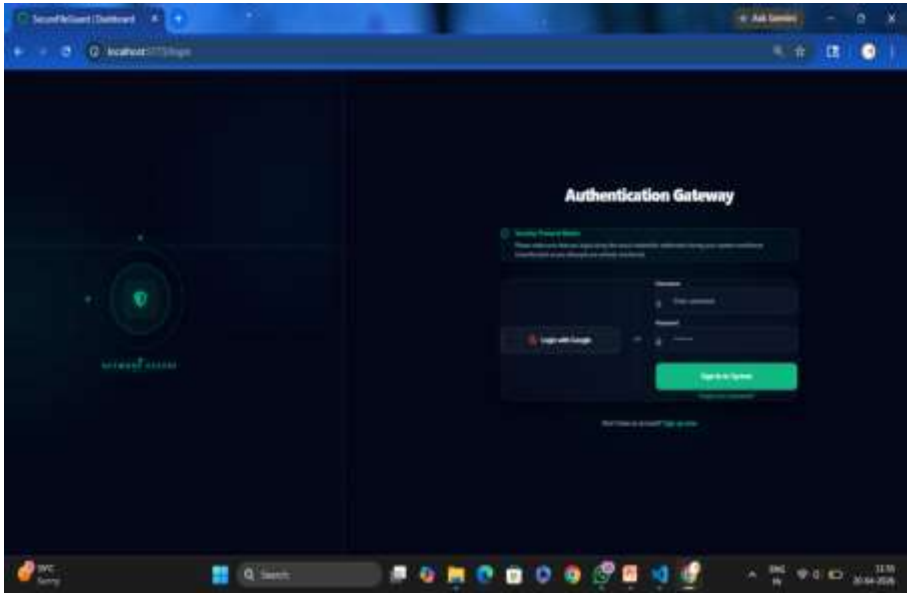
MODULE OUTPUT

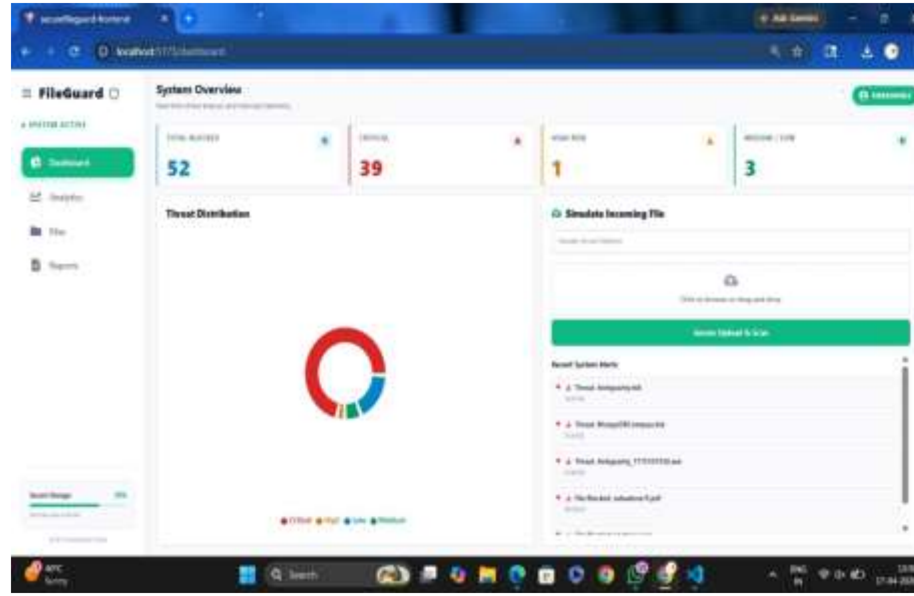
MODULE OUTPUT :





MODULE OUTPUT





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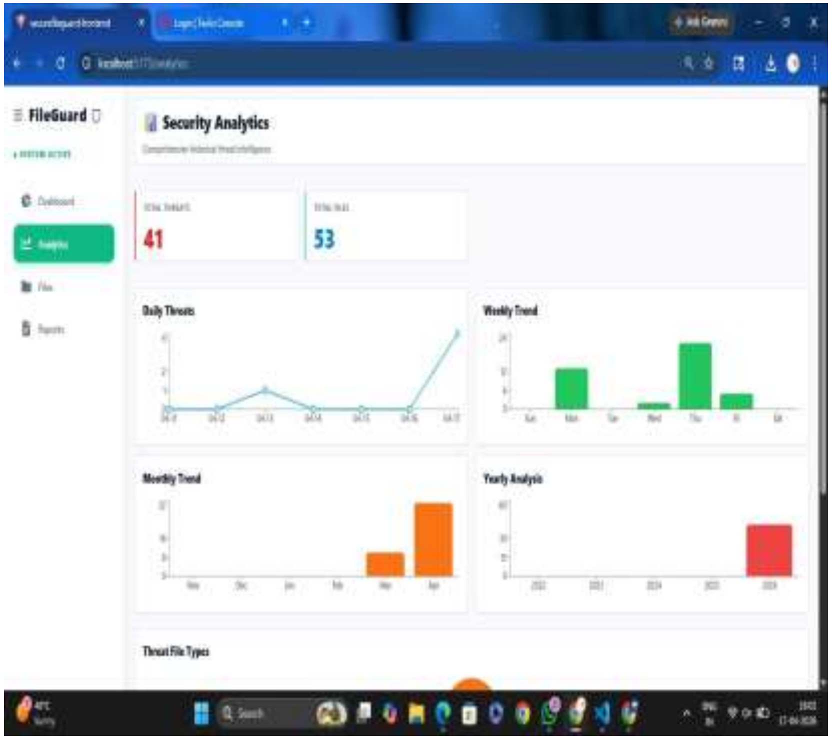

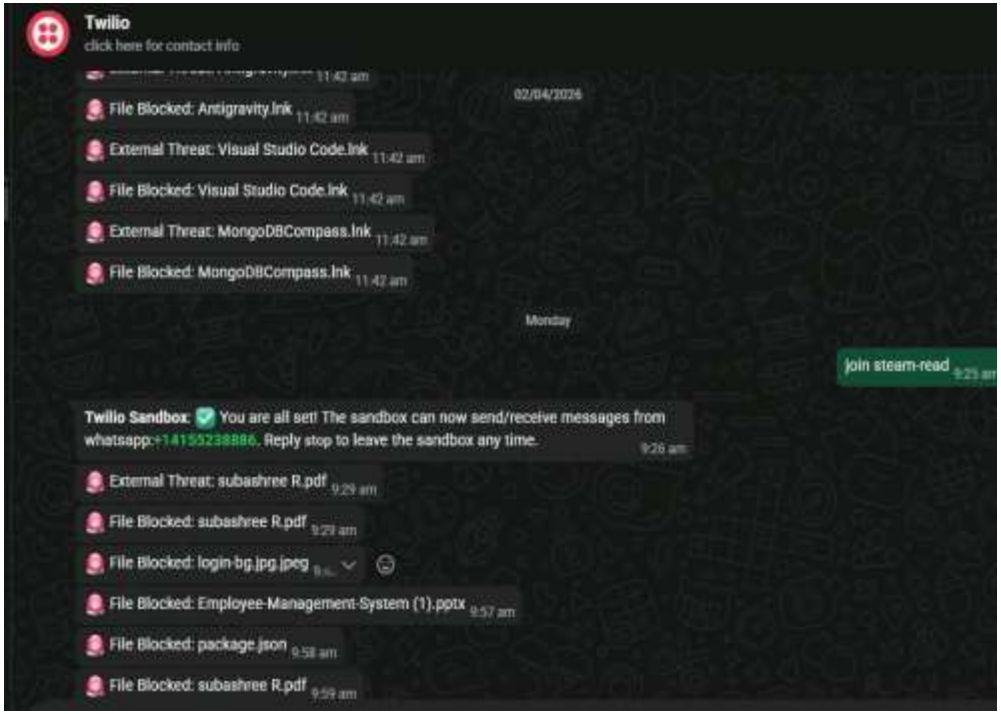

Conclusion

- Ransomware attacks are increasing rapidly and have become one of the biggest cybersecurity challenges today. Traditional security systems are no longer enough because modern ransomware keeps changing and becoming more advanced.

- In this project, we explored how machine learning–based intrusion detection systems can help identify ransomware attacks by analysing system behaviour, network activity, and unusual patterns.

- This intelligent approach makes it possible to detect threats early and reduce damage to data and systems.

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THANK YOU